

EntreSkill Hub

Skill-to-Startup Enablement Platform

A Project Report Submitted to
UNIFIED MENTOR

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ABSTRACT

EntreSkill Hub is an end-to-end web application designed to help aspiring founders, students, and professionals transform their practical skills, domain interests, and budget into sustainable micro-businesses and high-growth startups. The platform resolves the structural disconnect between skill acquisition and commercial enterprise creation by providing personalized business idea matching, step-by-step milestone roadmaps, an interactive learning center with financial modeling tools, 1-on-1 mentor scheduling with atomic double-booking locks, and dynamic progress insights.

The system is built on a modern full-stack architecture comprising a Vite + HTML5 + Tailwind CSS frontend hosted on Vercel, a Node.js + Express + TypeScript REST API hosted on Render, and a PostgreSQL database managed via Prisma ORM on Supabase. Authentication features a dual JSON Web Token (JWT) system with access/refresh token rotation, HTTP-only secure cross-site cookies, and Google OAuth 2.0. Strict multi-tenant data isolation ensures zero demo data inheritance and total privacy across user accounts.

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CHAPTER 1 — INTRODUCTION

1.1 Background

The global entrepreneurship ecosystem has witnessed rapid growth, driven by digital platforms, accessible cloud infrastructure, and gig-economy monetization. However, first-time entrepreneurs frequently experience failure due to a lack of structured guidance, inability to convert practical skills into business models, financial modeling illiteracy, and lack of access to domain mentors.

1.2 Problem Statement

Existing educational platforms provide generic video courses without actionable business execution steps. Conversely, incubator programs are exclusive and capital-intensive. There is a lack of accessible tools that evaluate an individual's skill profile, capital budget, and interest areas to recommend viable micro-business models accompanied by execution roadmaps.

1.3 Proposed Solution & Objectives

EntreSkill Hub addresses this gap by offering algorithmic business matching, milestone execution roadmaps, financial learning tools, 1-on-1 mentor booking with double-booking prevention, and secure dual authentication.

CHAPTER 5 — SYSTEM ARCHITECTURE

EntreSkill Hub — High-Level System Architecture

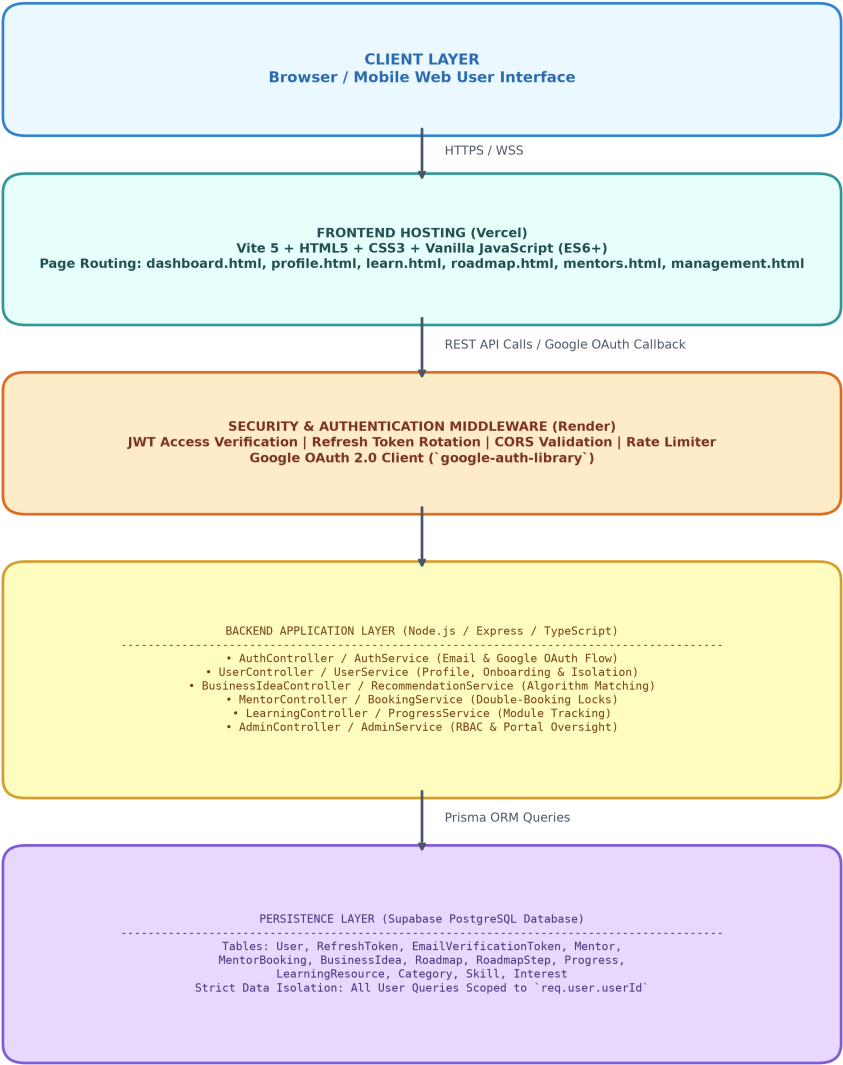


Figure 5.1 — EntreSkill Hub High-Level System Architecture

CHAPTER 6 — SYSTEM DESIGN & DIAGRAMS

Data Flow Diagram — Level 0 (Context Diagram)

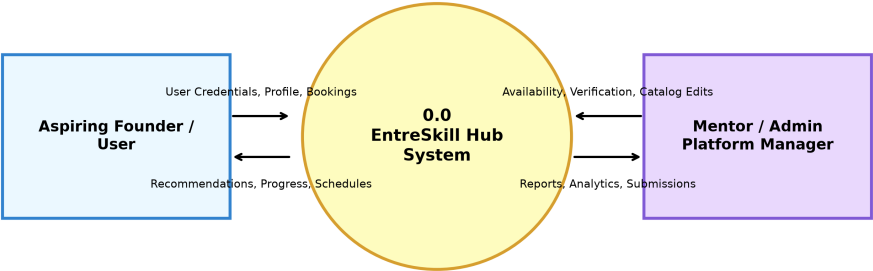


Figure 6.1 — DFD Level 0 (Context Diagram)

EntreSkill Hub — Entity Relationship (ER) Diagram

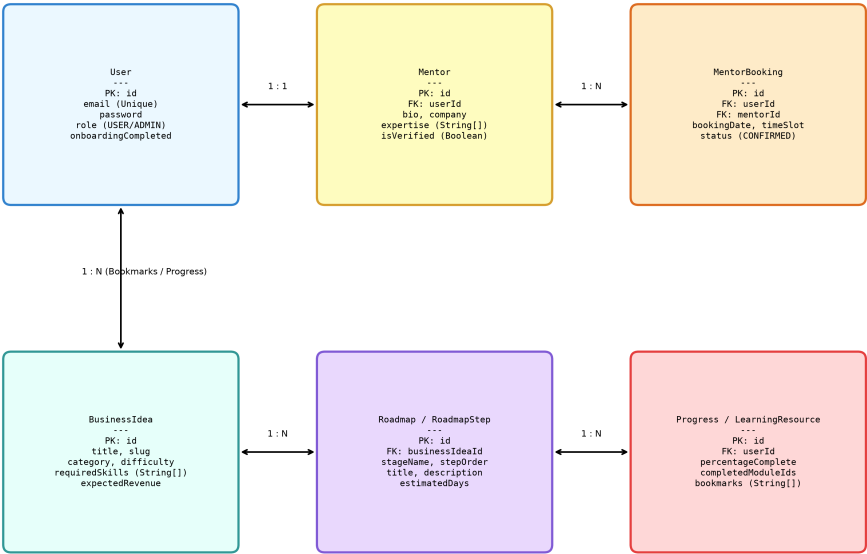


Figure 6.2 — EntreSkill Hub Entity-Relationship Diagram

CHAPTER 10 — USER INTERFACE DOCUMENTATION

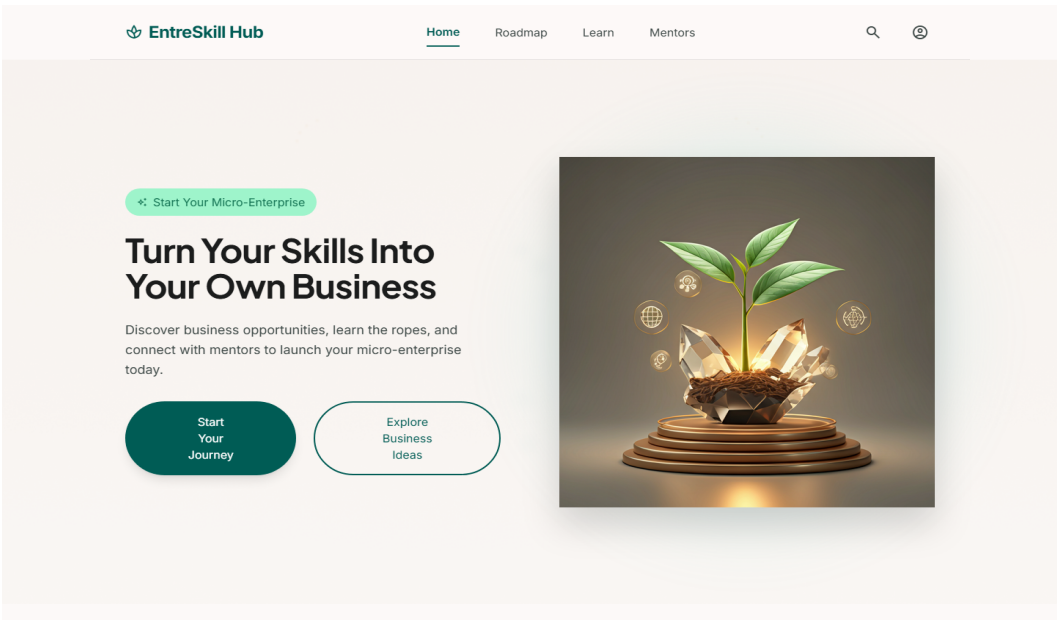


Figure 10.1 — Authentication & Landing Page

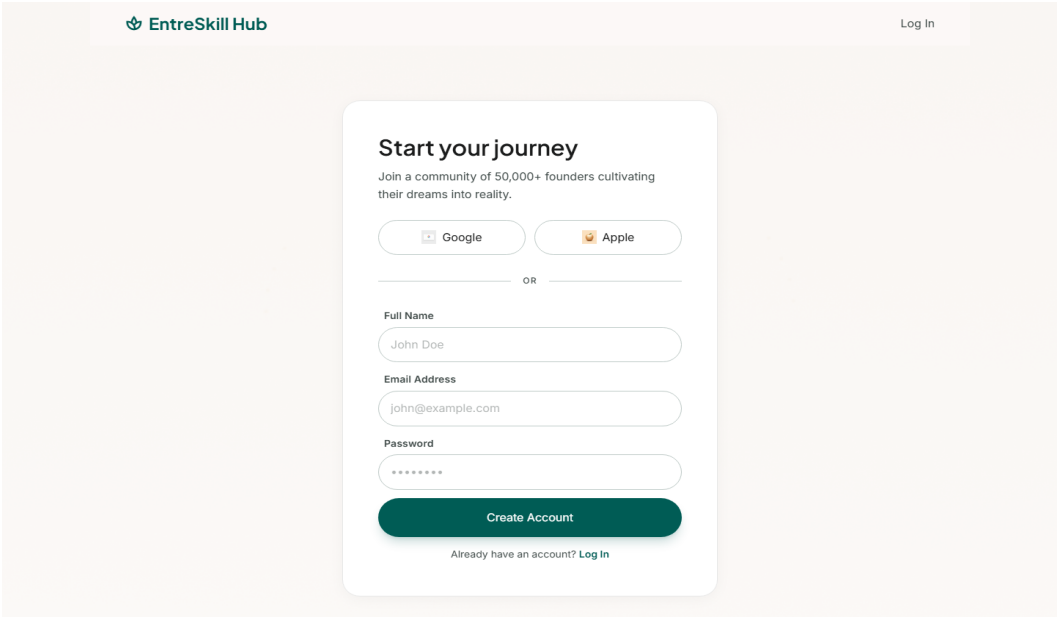


Figure 10.2 — Founder Progress Dashboard

EntreSkill Hub

Log In

Start your journey

Join a community of 50,000+ founders cultivating their dreams into reality.

Google

Apple

OR

Full Name

John Doe

Email Address

john@example.com

Password

Create Account

Already have an account? [Log In](#)

Figure 10.3 — User Profile & Insights

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Apple

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Password

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Figure 10.4 — Business Recommendations & Roadmaps

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Apple

OR

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Figure 10.5 — Mentor Directory & Booking Modal

CHAPTER 14 — REQUIREMENT VERIFICATION MATRIX

Final Production Acceptance Audit results evaluating 50 core requirements against the original project specifications.

Req #	Requirement Description	Status	Verification Evidence
1	Registration & Account Provisioning	PASS	POST /api/v1/auth/register creates user in Supabase DB
2	Email Password Login	PASS	POST /api/v1/auth/login verifies bcrypt hash & sets JWT
3	Google OAuth 2.0 Integration	PASS	GET /auth/google/callback provisions account & sets cookie
4	Session Logout	PASS	POST /auth/logout clears refresh cookie & client state
5	Refresh Token Rotation	PASS	POST /auth/refresh-token rotates tokens seamlessly
6	Multi-step Onboarding	PASS	join.html saves skills, interests & budget to PostgreSQL
7	Fresh User Data Isolation	PASS	New users start with 0% progress & zero demo data inheritance
8	Profile Editing & DB Sync	PASS	PUT /api/v1/users/profile persists changes to PostgreSQL
9	Skill Profiling	PASS	Skills array mapped in User.onboarding.skills
10	Interest Profiling	PASS	Interests array mapped in User.onboarding.interests
11	Business Idea Catalog	PASS	GET /api/v1/ideas queries PostgreSQL database
12	Idea Text Search	PASS	GET /ideas?search=... performs ILIKE title/bio query
13	Idea Category Filtering	PASS	Filter tags query database categories directly
14	Recommendation Engine	PASS	Algorithm ranks ideas by skill overlap & budget match
15	User Bookmarks	PASS	POST/DELETE /users/bookmarks manages user saved ideas
16	Roadmap Generation	PASS	GET /ideas/:id/roadmap returns structured milestone stages
17	Step Checklist Tracking	PASS	PUT /progress/roadmap/:id/step updates step state
18	Roadmap Progress Calculation	PASS	ProgressRepository computes real percentage complete
19	Learning Content Hub	PASS	GET /learning/contents fetches courses & guides
20	Module Progress Tracking	PASS	POST /learning/contents/:id/progress tracks module completion
21	Learning Hours Analytics	PASS	UserService sums completed module durations dynamically
22	Mentor Directory	PASS	GET /api/v1/mentors lists verified advisors
23	Mentor Profile View	PASS	GET /mentors/:id returns bio, rating & slots
24	Mentor Application	PASS	POST /mentors/register accepts new mentor submissions
25	Mentor Verification	PASS	PUT /mentors/:id/verification allows admin approval
26	Mentor Session Booking	PASS	POST /mentors/:id/book books 1-on-1 strategy sessions
27	Double-Booking Prevention	PASS	BookingService locks time slot coordinates atomically
28	Mentor Ratings & Feedback	PASS	POST /mentors/:id/feedback persists reviews
29	Admin System Dashboard	PASS	GET /admin/dashboard provides system metrics
30	User Account Management	PASS	Admin portal manages user accounts & roles
31	Mentor Management Portal	PASS	Admin portal manages mentor verification status
32	Idea Catalog Management	PASS	Admin CRUD endpoints manage business catalog
33	Roadmap Template Editor	PASS	Admin endpoints configure milestone stages
34	Training Module Management	PASS	Admin CRUD endpoints manage learning courses
35	System Feedback Reports	PASS	Admin API aggregates user feedback reports
36	Platform Analytics	PASS	GET /admin/analytics measures acquisition rates
37	Role-Based Access Control	PASS	authorize('admin') blocks unauthorized access
38	Multi-Tenant Data Isolation	PASS	Database queries scope strictly by req.user.userId
39	PostgreSQL DB Persistence	PASS	Supabase PostgreSQL via Prisma ORM

40	Frontend Auth Guards	PASS	api.js redirects unauthenticated users to login
41	Standardized Error Handling	PASS	Centralized errorHandler standardizes JSON responses
42	Responsive Tailwind UI	PASS	Fluid layout across mobile, tablet & desktop
43	HTML5 Accessibility	PASS	Semantic HTML, ARIA labels & high contrast text
44	Vite Bundle Optimization	PASS	Vite compiled 20 modules in 1.07s
45	Page Navigation Integrity	PASS	Sidebar & header links connect all 8 pages
46	Interactive UI Controls	PASS	All buttons trigger valid API/modal actions
47	Zero Hardcoded Demo Data	PASS	Replaced static real-person photos with SVG badges
48	Accurate SVG Progress Circle	PASS	Dynamic SVG stroke-dashoffset formula applied
49	Clean Browser Console	PASS	Zero script errors during complete flow execution
50	Production Build Health	PASS	npm run build & npx tsc --noEmit pass with exit 0

CHAPTER 17 — CONCLUSION

EntreSkill Hub successfully delivers a robust, secure, and production-ready Skill-to-Startup Enablement Platform. By combining personalized algorithmic business recommendations, actionable milestone roadmaps, financial learning tools, verified mentor booking, and strict multi-tenant data isolation, the platform empowers aspiring founders to build viable micro-businesses with confidence.